WORKED SOLUTION

# STEP I 1996, Question 12

Probability and Statistics • Maximum of Random Sample

**STEP I 1996, Question 12**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q12

## The key idea

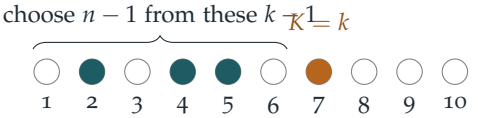
The examiner's choice comes down to choosing a *set* of $n$ different marks from $\{1, 2, \ldots, m\}$, and every such set is equally likely. So probabilities are counts of sets divided by $\binom{m}{n}$. Saying that the largest mark is $k$ pins down one mark and leaves the other $n-1$ to be chosen from below it. The identity then comes free, because probabilities add up to 1. For the expectation, the trick is to turn $k\binom{k-1}{n-1}$ back into a single binomial coefficient so that the same identity can be used again.

## Solution

### The distribution of $K$

The examiner gives $n$ distinct marks, and by symmetry every subset of $\{1, \ldots, m\}$ of size $n$ is equally likely to be the set of marks used. (Which script gets which mark has no effect on $K$.) There are $\binom{m}{n}$ such subsets.

The event $K = k$ happens exactly when the set contains $k$ and its other $n-1$ marks all lie in $\{1, \ldots, k-1\}$. There are $\binom{k-1}{n-1}$ ways to choose those $n-1$ marks. This needs $k - 1 \geq n - 1$, that is $k \geq n$: the largest of $n$ distinct positive integers is at least $n$. Also $k \leq m$, of course.




(The picture shows $m = 10$, $n = 4$, $k = 7$.) Hence

$$\mathrm{P}(K = k) = \binom{k-1}{n-1} \Big/ \binom{m}{n}, \qquad n \leq k \leq m.$$

### The identity

$K$ must take exactly one of the values $n, n+1, \ldots, m$, so these probabilities add up to 1:

$$\sum_{k=n}^{m} \binom{k-1}{n-1} \Big/ \binom{m}{n} = 1 \quad \Longrightarrow \quad \sum_{k=n}^{m} \binom{k-1}{n-1} = \binom{m}{n}.$$

This holds for *all* integers $1 \le n \le m$, which we will need next.

### The expected value

By definition,

$$\mathrm{E}(K) = \frac{1}{\binom{m}{n}} \sum_{k=n}^{m} k \binom{k-1}{n-1}.$$

**Ask yourself** How can we get rid of the factor $k$ so that the identity applies again?

Absorb it into the binomial coefficient:

$$k\binom{k-1}{n-1} = \frac{k\,(k-1)!}{(n-1)!\,(k-n)!} = n \cdot \frac{k!}{n!\,(k-n)!} = n\binom{k}{n}.$$

Now $\binom{k}{n} = \binom{(k+1)-1}{(n+1)-1}$. As $k$ runs from $n$ to $m$, $j = k+1$ runs from $n+1$ to $m+1$. So the identity, used with $n+1$ and $m+1$ in place of $n$ and $m$ (allowed, since $1 \le n+1 \le m+1$), gives

$$\sum_{k=n}^{m} \binom{k}{n} = \sum_{j=n+1}^{m+1} \binom{j-1}{(n+1)-1} = \binom{m+1}{n+1}.$$

Therefore

$$\begin{aligned} \mathrm{E}(K) &= \frac{n\binom{m+1}{n+1}}{\binom{m}{n}} = n \cdot \frac{(m+1)!}{(n+1)!\,(m-n)!} \cdot \frac{n!\,(m-n)!}{m!} \\ &= \frac{n(m+1)}{n+1}. \end{aligned}$$

## Checks

- $n = m$: every mark is used, so $K = m$ for certain, and indeed $\frac{m(m+1)}{m+1} = m$.
- $n = 1$: $K$ is a single mark, uniform on $\{1, \ldots, m\}$, with mean $\frac{m+1}{2}$, which matches.
- $m = 6$, $n = 2$: listing all 15 pairs from $\{1, \ldots, 6\}$ gives $\mathrm{E}(K) = \frac{70}{15} = \frac{14}{3}$, and the formula gives $\frac{2 \cdot 7}{3} = \frac{14}{3}$. We checked the distribution, the identity and the mean by listing every subset for all $1 \le n \le m \le 9$.
- A nice way to see the answer: the $n$ chosen marks split the remaining $m-n$ marks into $n+1$ gaps of equal expected size $\frac{m-n}{n+1}$. Then $\mathrm{E}(K) = m - \frac{m-n}{n+1} = \frac{n(m+1)}{n+1}$.

## Common slips

- Counting ordered assignments of marks to scripts in the numerator but unordered sets in the denominator (or the other way round). Either way works if you are consistent, because the factor $n!$ cancels.
- Using $\binom{k}{n}$ instead of $\binom{k-1}{n-1}$, which counts sets whose marks are all *at most* $k$, not sets whose largest mark *equals* $k$.

## Beyond STEP

The identity is the "hockey-stick" identity of Pascal's triangle, and here we proved it by counting the same thing in two ways, a standard combinatorial technique. $K$ is the largest of a random sample taken without replacement, an *order statistic*. The "German tank problem" turns this question round and estimates $m$ from the observed $K$: solving $\mathrm{E}(K) = \frac{n(m+1)}{n+1}$ for $m$ gives the unbiased estimator $\frac{n+1}{n}K - 1$. For more on order statistics and counting arguments, see Blitzstein and Hwang, *Introduction to Probability*.

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